Control Systems
Cheat Sheet

Written by

Andrea Covelli

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Introduction

This document serves as a comprehensive cheat sheet for control systems, covering essential topics such as root locus plots, Nyquist diagrams, and lead compensator design. It includes theoretical foundations, mathematical properties, and step-by-step procedures to aid in the analysis and design of control systems.

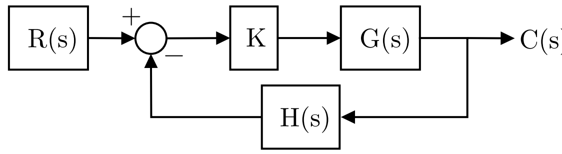
Note

This cheat sheet is written by a student and may contain errors or inaccuracies. It is intended as a study aid and should not be used as a primary source of information. Please consult textbooks, lecture notes, or other reliable sources for critical information.

1 Rules for Making Root Locus Plots

The closed-loop transfer function of the system shown is:

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$



So the characteristic equation (c.e.) is

$$1 + KG(s)H(s) = 1 + K\frac{N(s)}{D(s)} = 0, \text{ or } D(s) + KN(s) = 0$$

As K changes, so do the locations of the closed-loop poles (i.e., zeros of the characteristic equation). The table below gives rules for sketching the location of these poles for $K \in [0, +\infty)$ (i.e., $K \geq 0$).

Rule Name	**Description**
Definitions	• The loop gain is $KG(s)H(s)$ or $K\frac{N(s)}{D(s)}$ • $N(s)$, the numerator, is an m^{th}-order polynomial; $D(s)$ is an n^{th}-order polynomial • $N(s)$ has zeros at z_i $(i = 1..m)$; $D(s)$ has zeros (which are the loop-gain poles) at p_i $(i = 1..n)$ • The difference between n and m is q, so $q = n - m$ $(q \geq 0)$
Symmetry	The locus is symmetric about the real axis (i.e., complex poles appear as conjugate pairs)
Number of Branches	There are n branches of the locus, one for each closed-loop pole
Starting and Ending Points	The locus starts (at $K = 0$) at the poles of the loop gain, and ends (as $K \to +\infty$) at the zeros. Note: This means that there will be q branches that go to infinity as $K \to +\infty$
Locus on Real Axis*	For any point s on the real axis to be a point on the root locus, the sum of the number of open-loop poles and zeros of $G(s)H(s)$ to the right of that point must be odd
Asymptotes as $\|s\| \to \infty$*	If $q > 0$, there are asymptotes of the root locus that intersect the real axis at $$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{q}$$ and radiate out with angles $$\theta = \pm r\frac{180^\circ}{q}, \text{ where } r = 1, 3, 5, ...$$
Break-Away/-In Points	Break-away or break-in points of the locus exist where $N(s)D'(s) - N'(s)D(s) = 0$. Note: Only solutions yielding a real and non-negative gain $K \geq 0$ lie on the root locus.

Rule Name	Description
Angle of Departure*	Angle of departure from a pole p_j is $$\theta_{depart,p_j} = 180^\circ + \sum_{i=1}^{m} \angle(p_j - z_i) - \sum_{i=1,i\neq j}^{n} \angle(p_j - p_i)$$
Angle of Arrival*	Angle of arrival at a zero z_j is $$\theta_{arrive,z_j} = 180^\circ - \sum_{i=1,i\neq j}^{m} \angle(z_j - z_i) + \sum_{i=1}^{n} \angle(z_j - p_i)$$
Imaginary Axis Crossing	Use Routh-Hurwitz to determine where the locus crosses the imaginary axis
Finding Poles for Given K	Rewrite the characteristic equation (c.e.) as $D(s) + KN(s) = 0$. Substitute the value of K into the equation and find the roots of the characteristic equation. (This may require a computer)
Finding K for Given Pole	Rewrite the c.e. as $K = -\frac{D(s)}{N(s)}$, replace s with the desired pole location and solve for K. Note: If s is not exactly on the locus, K may be complex (with a small imaginary part). Use the real part of K.

*These rules change to draw complementary root locus ($K \leq 0$). See below for details.

2 Complementary Root Locus

To sketch the complementary root locus ($K \leq 0$, also known as the *inverse root locus*), most of the rules are unchanged except for those in the table below.

Rule Name	Description
Locus on Real Axis	For any point s on the real axis to be a point on the complementary root locus, the sum of the number of open-loop poles and zeros of $G(s)H(s)$ to the right of that point must be even
Asymptotes	If $q > 0$, there are asymptotes of the complementary root locus that intersect the real axis at $$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{q}$$ and radiate out with angles $$\theta = \pm p\frac{180^\circ}{q}, \text{ where } p = 0, 2, 4, ...$$

Rule Name	Description
Angle of Departure	Angle of departure from a pole p_j is $$\theta_{depart,p_j} = \sum_{i=1}^{m} \angle(p_j - z_i) - \sum_{i=1, i\neq j}^{n} \angle(p_j - p_i)$$
Angle of Arrival	Angle of arrival at a zero z_j is $$\theta_{arrive,z_j} = -\sum_{i=1, i\neq j}^{m} \angle(z_j - z_i) + \sum_{i=1}^{n} \angle(z_j - p_i)$$

3 The Nyquist Diagram: Theoretical Foundations

3.1 Complex Analysis Foundations

The Nyquist diagram is rooted in complex analysis. For a transfer function $G(s)$, we consider its behavior along a contour in the complex plane. The fundamental theorem underlying the Nyquist plot is Cauchy's argument principle:

$$Z = N + P$$

where:

- Z = number of zeros of $1 + G(s)H(s)$ enclosed by the contour (corresponding to closed-loop poles in the right half-plane)
- N = number of clockwise encirclements of the critical point $-1 + j0$ by the Nyquist plot of $G(s)H(s)$
- P = number of poles of $G(s)H(s)$ enclosed by the contour (corresponding to open-loop poles in the right half-plane)

3.2 Phase and Gain Margins

From the Nyquist plot of the loop transfer function $G(s)H(s)$:

- Phase Margin: $\phi_m = \angle G(j\omega_{gc})H(j\omega_{gc}) + 180^\circ$
- Gain Margin: $G_m = \dfrac{1}{|G(j\omega_{pc})H(j\omega_{pc})|}$
  In decibels: $G_{m,\text{dB}} = 20\log_{10} G_m = -20\log_{10}|G(j\omega_{pc})H(j\omega_{pc})|$

where:

- ω_{gc} is the gain crossover frequency where $|G(j\omega_{gc})H(j\omega_{gc})| = 1$
- ω_{pc} is the phase crossover frequency where $\angle G(j\omega_{pc})H(j\omega_{pc}) = -180^\circ$

3.3 Transfer Function Analysis

Given a rational transfer function:

$$G(s) = K\frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{k=1}^{n}(s - p_k)}$$

where:

- K is the gain
- z_i are zeros
- p_k are poles

The frequency response is obtained by evaluating $G(s)$ along $s = j\omega$:

$$G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)}$$

3.4 Mathematical Properties

3.4.1 Magnitude Calculation

For any frequency ω:

$$|G(j\omega)| = |K|\frac{|j\omega - z_1||j\omega - z_2|\cdots|j\omega - z_m|}{|j\omega - p_1||j\omega - p_2|\cdots|j\omega - p_n|}$$

3.4.2 Phase Calculation

The phase is given generally by:

$$\begin{aligned}\angle G(j\omega) &= \angle K + \angle(j\omega - z_1) + \angle(j\omega - z_2) + \cdots + \angle(j\omega - z_m) \\ &\quad - \angle(j\omega - p_1) - \angle(j\omega - p_2) - \cdots - \angle(j\omega - p_n) \\ &= \angle K + \sum_{i=1}^{m} \angle(j\omega - z_i) - \sum_{k=1}^{n} \angle(j\omega - p_k)\end{aligned}$$

For real poles and zeros located in the left half-plane ($z_i = -\sigma_{z,i}$, $p_k = -\sigma_{p,k}$ with $\sigma_{z,i}, \sigma_{p,k} > 0$), we have:

$$\angle(j\omega - z_i) = \tan^{-1}\left(\frac{\omega}{\sigma_{z,i}}\right), \quad \angle(j\omega - p_k) = \tan^{-1}\left(\frac{\omega}{\sigma_{p,k}}\right)$$

and any pole at the origin of multiplicity n_0 contributes $-n_0 \cdot 90^\circ$ to the phase.

3.5 Strictly Proper Functions

For strictly proper functions ($n > m$):

$$\lim_{\omega\to\infty} G(j\omega) = 0$$

$$\lim_{\omega\to\infty} |G(j\omega)| = \lim_{\omega\to\infty} |K|\frac{\omega^m}{\omega^n} = 0$$

4 Nyquist Plot Construction

The construction of the Nyquist plot is divided into three main sections, each corresponding to different portions of the Nyquist contour.

4.1 General Procedure Overview

For any transfer function $G(s)$, the Nyquist plot is constructed by following these steps in order:

Step 0. Compute $G(j\omega)$ and $\angle G(j\omega)$. Do not simplify the numerator/denominator because they will be used to find the signs of the real and imaginary parts of $G(j\omega)$ in Step 2.

Step 1. First Arc: $\omega \to 0^+$, $\theta \in [0^\circ, 90^\circ]$

Let $s = \varepsilon e^{j\theta}$, where $\varepsilon \to 0$ (the radius of a small detour around a pole at the origin of multiplicity n_0). As $s \to 0$, $G(s)$ behaves asymptotically as:

$$G(s) \approx \frac{C}{\varepsilon^{n_0}} e^{-jn_0\theta}$$

where C is a constant.

- Move along the arc from $\theta = 0^\circ$ to $\theta = 90^\circ$
- Magnitude stays at ∞ as $\varepsilon \to 0$
- Substitute θ values into the asymptotic form to find $\angle G(s)$

	$\lvert G(s)\rvert$	$\angle G(s)$
$\theta_{\mathrm{i}} = 0^\circ$	∞	$\angle G(\varepsilon e^{j\theta})\vert_{\theta=0^\circ}$
$\theta_{\mathrm{f}} = 90^\circ$	∞	$\angle G(\varepsilon e^{j\theta})\vert_{\theta=90^\circ}$

Step 2. Imaginary Axis Traversal: $\omega \in (0^+, +\infty)$, $\theta = 90^\circ$

- Start at the end of the first arc
- Move along the imaginary axis with $s = j\omega$
- The angle of the contour variable s is maintained at 90° throughout

	$\lvert G(j\omega)\rvert$	$\Re\{G(j\omega)\}$	$\Im\{G(j\omega)\}$	$\angle G(j\omega)$
$\omega_{\mathrm{i}} = 0^+$	$\lvert G(j\omega_{\mathrm{i}})\rvert$	$\Re\{G(j\omega_{\mathrm{i}})\}$	$\Im\{G(j\omega_{\mathrm{i}})\}$	$\angle G(j\omega_{\mathrm{i}})$
$\omega_{\mathrm{f}} = +\infty$	$\lvert G(j\omega_{\mathrm{f}})\rvert$	$\Re\{G(j\omega_{\mathrm{f}})\}$	$\Im\{G(j\omega_{\mathrm{f}})\}$	$\angle G(j\omega_{\mathrm{f}})$

- Note 1: $\Re\{G(j\omega)\}$ and $\Im\{G(j\omega)\}$ are respectively the real and imaginary parts of $G(j\omega)$
- Note 2: $\angle G(j\omega)$ is the phase of $G(j\omega)$

Step 3. Infinite Radius Arc: $R \to +\infty$, $\theta \in [90^\circ, 0^\circ]$

Let $s = Re^{j\theta}$, where $R \to +\infty$. As $R \to +\infty$, $G(s)$ behaves asymptotically as:

$$G(s) \approx \frac{C}{R^q} e^{-jq\theta}$$

where $q = n - m$ is the relative degree of the transfer function, and C is a constant.

- Move along the arc from $\theta = 90^\circ$ to $\theta = 0^\circ$
- Magnitude stays at 0 as $R \to +\infty$

- Substitute θ values into the asymptotic form to find $\angle G(s)$

	$\|G(s)\|$	$\angle G(s)$
$\theta_{\mathrm{i}} = 90^\circ$	0	$\angle G(Re^{j\theta})\|_{\theta=90^\circ}$
$\theta_{\mathrm{f}} = 0^\circ$	0	$\angle G(Re^{j\theta})\|_{\theta=0^\circ}$

Draw the Nyquist plot by connecting the points obtained in each step.

Considering the symmetry of the Nyquist plot, once the plot is completed for $\omega \in (0^+, +\infty)$, the plot must be mirrored with respect to the real axis to obtain the plot for $\omega \in (-\infty, 0^-)$.

5 Special Cases

5.1 Systems Without Poles on the $j\omega$-axis

For systems without poles on the imaginary axis, the following simplifications apply:

- Step 1 is omitted because no detour around the origin (or any other point on the imaginary axis) is needed.
- In Step 2, the imaginary axis traversal starts exactly at $\omega_i = 0$ instead of 0^+.
- Step 3 is evaluated simply to find the terminal point of the plot as $R \to +\infty$.

6 Worked Example

Consider the transfer function:

$$G(s) = \frac{K}{s(s+2)}$$

Let's construct the Nyquist plot following our procedure:

Step 0: Initial Analysis

Substitute $s = j\omega$:

$$G(j\omega) = \frac{K}{j\omega(j\omega+2)} = \frac{K}{-\omega^2 + 2j\omega}$$

Multiply the numerator and denominator by the complex conjugate:

$$G(j\omega) = \frac{K(-\omega^2 - 2j\omega)}{(-\omega^2)^2 + (2\omega)^2} = \frac{-K\omega^2}{4\omega^2+\omega^4} + j\frac{-2K\omega}{4\omega^2+\omega^4}$$

By simplifying, we obtain:

$$\Re\{G(j\omega)\} = -\frac{K}{\omega^2+4}$$

$$\Im\{G(j\omega)\} = -\frac{2K}{\omega(\omega^2+4)}$$

Phase:

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{2}\right)$$

Step 1: First Arc ($\omega \to 0^+$, $\theta \in [0^\circ, 90^\circ]$)

For $s = \varepsilon e^{j\theta}$:

$$G(s) = \frac{K}{\varepsilon e^{j\theta}(\varepsilon e^{j\theta}+2)} = \frac{K}{\varepsilon^2 e^{2j\theta} + 2\varepsilon e^{j\theta}}$$

As $\varepsilon \to 0$, $G(s)$ behaves asymptotically as:

$$G(s) \approx \frac{K}{2\varepsilon} e^{-j\theta} \to \infty$$

Then we have:

	$\lvert G(s)\rvert$	$\angle G(s)$
$\theta_{\mathrm{i}} = 0^\circ$	$+\infty$	0°
$\theta_{\mathrm{f}} = 90^\circ$	$+\infty$	-90°

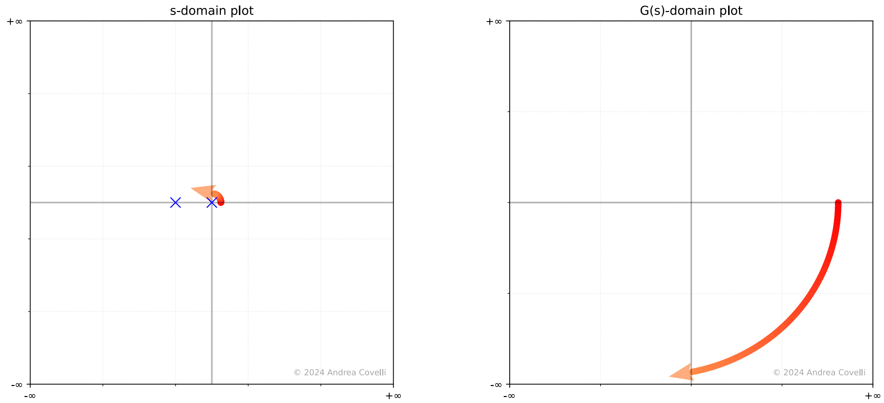


Figure 1: The image is a numerical approximation of the Nyquist plot; the direction of the arc may vary slightly.

Step 2: Imaginary Axis Traversal ($\omega \in (0, +\infty)$)

Key points along the imaginary axis:

ω	$\Re\{G(j\omega)\}$	$\Im\{G(j\omega)\}$	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
0^+	$-K/4$	$-\infty$	$+\infty$	-90°
$+\infty$	0^-	0^-	0^+	-180°

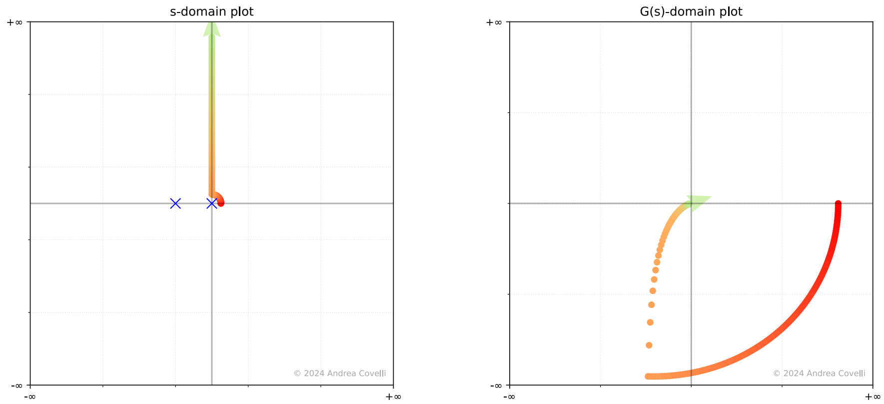


Figure 2: The image is a numerical approximation of the Nyquist plot; the direction of the arc may vary slightly.

Step 3: Infinite Radius Arc ($R \to +\infty, \theta \in [90^\circ, 0^\circ]$)

For $s = Re^{j\theta}$ as $R \to +\infty$, $G(s)$ behaves asymptotically as:

$$G(s) \approx \frac{K}{R^2} e^{-2j\theta} \to 0$$

Then we have:

	$\lvert G(s)\rvert$	$\angle G(s)$
$\theta_\mathrm{i} = 90^\circ$	0^+	-180°
$\theta_\mathrm{f} = 0^\circ$	0^+	0°

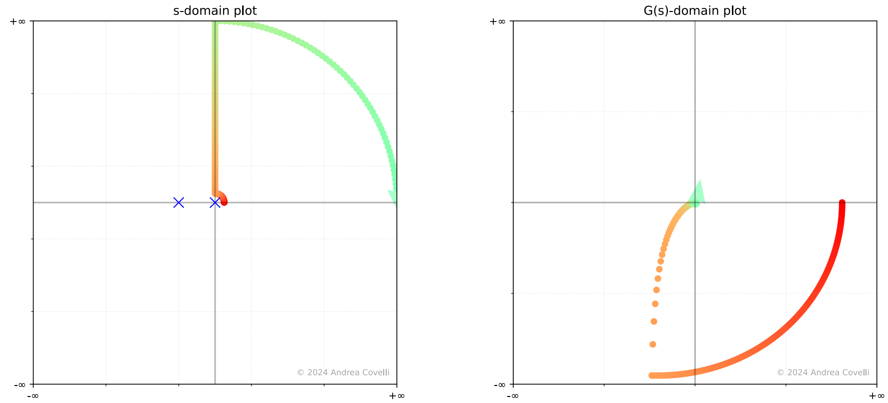


Figure 3: The image is a numerical approximation of the Nyquist plot; the direction of the arc may vary slightly.

Final Plot

The Nyquist plot is obtained by connecting the points obtained in each step. The plot is then mirrored with respect to the real axis to obtain the plot for $\omega \in (-\infty, 0^-)$.

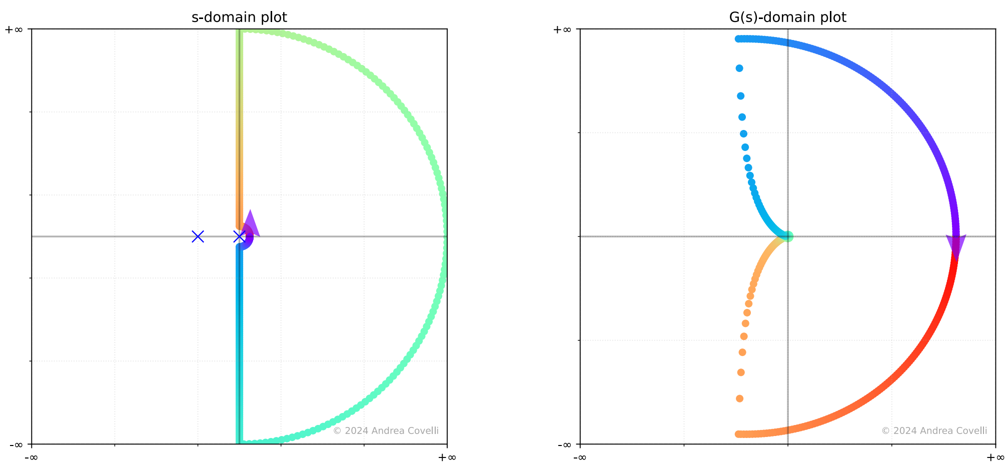


Figure 4: The image is a numerical approximation of the Nyquist plot; the direction of the arc may vary slightly.

According to the Nyquist stability criterion, since the open-loop system has no poles in the right half-plane ($P = 0$) and the Nyquist plot does not encircle the critical point $-1 + j0$ ($N = 0$), the number of closed-loop poles in the right half-plane is $Z = N + P = 0$.

Therefore, the closed-loop system is stable.

7 Common Mathematical Pitfalls

- Sign errors in phase calculations:

$$\angle(a + jb) = \arctan\left(\frac{b}{a}\right) + \begin{cases} 0^\circ & \text{if } a > 0 \\ 180^\circ & \text{if } a < 0 \end{cases}$$

- Rotation direction in the Nyquist diagram:

For $\Delta\phi = \phi_f - \phi_i$ (where $\phi = \angle G(s)$), the rotation of the Nyquist plot is:

$$\begin{cases} \text{counter-clockwise} & \text{if } \Delta\phi > 0 \\ \text{clockwise} & \text{if } \Delta\phi < 0 \end{cases}.$$

8 Lead Compensator Design

8.1 Introduction

Lead compensators are used to improve the transient response of a system. They are designed to increase the phase margin of a system, which in turn increases the stability of the system.

The transfer function of a lead compensator is given by:

$$D(s) = K\frac{T_d s + 1}{\alpha T_d s + 1}, \quad \alpha < 1$$

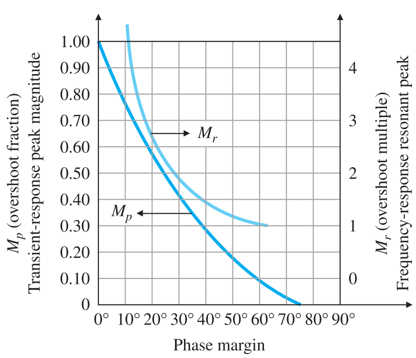
where:

- K is the DC gain of the compensator (often determined by steady-state error constraints)
- $T_d = \dfrac{1}{\omega_{gc}\sqrt{\alpha}}$ where ω_{gc} is the desired gain crossover frequency at which the maximum phase lead is placed
- α is the zero-pole ratio, $\alpha = \dfrac{|z|}{|p|}$

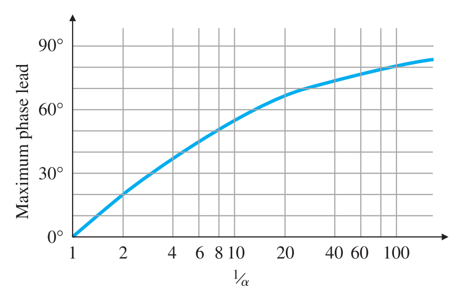
From the transfer function, the zero and pole of the compensator are given by:

- Zero: $s = -\dfrac{1}{T_d}$
- Pole: $s = -\dfrac{1}{\alpha T_d}$

Transient-response overshoot (M_p) and frequency-response resonant peak (M_r) versus PM for $T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$



Maximum phase increase for lead compensation



8.2 Design Procedure

To design a lead compensator, follow these steps:

1. Determine the compensator gain K to satisfy steady-state error requirements.
2. Evaluate the phase margin of the uncompensated system with gain K applied, i.e., $KG(s)$, and determine the required phase margin improvement $\phi_{\max}$ (including a safety margin of 5° to 12° to account for the shift in the crossover frequency).
3. Calculate the zero-pole ratio $\alpha < 1$ using the relation $\alpha = \dfrac{1 - \sin\phi_{\max}}{1 + \sin\phi_{\max}}$ (or estimate α from the maximum phase plots).
4. Select the desired new crossover frequency ω_{gc} such that the uncompensated system with gain K satisfies $|KG(j\omega_{gc})| = \sqrt{\alpha}$ (which corresponds to $20\log_{10}|KG(j\omega_{gc})| = 10\log_{10}\alpha$ dB).

5. Compute the time constant $T_d = \dfrac{1}{\omega_{gc}\sqrt{\alpha}}$ so that the peak phase lead occurs at ω_{gc}.
6. Place the zero at $s = -\dfrac{1}{T_d}$ and the pole at $s = -\dfrac{1}{\alpha T_d}$.
7. Verify that the phase margin has increased and all design criteria are met.